MIL-C-3883A

25 September 1970

SUPERSEDING
MIL-C-3883
18 May 1953

MILITARY SPECIFICATION

CORD, ELECTRICAL, SHIELDED (AUDIO FREQUENCY)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical cords composed of 1 to 7 conductors, 20 AWG, styrene butadiene rubber insulated and jacketed, with a shield beneath the jacket. These cords are intended for use in audio frequency applications, at temperatures down to -55°C. (See 3.1 for individual types.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-W-343	- Wire, Electrical (Uninsulated).

Military

MIL-C-572	- Cords, Yarn and Monofilaments, Organic Synthetic Fiber.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Non-Rigid.
MIL-I-3930	- Insulating and Jacketing Compounds, Electrical (For Cable, Cord, and Wire).
MIL-C-12000	- Cable, Cord and Wire, Electric; Packaging and Packing For.
MIL-C-45662	- Calibration System Requirements.

FSC 6145

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STANDARDS

Federal

FED-STD-228 - Federal Test Method Standard; Cable and Wire, Insulated; Method of Testing.

Military

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The cord types covered by this specification, including cord dimensions and electrical requirements, shall be as specified in table I.

Table I. Cord types, dimensions and electrical requirements.

Cord types	Number of conductors	Insulation thickness, minimum	Jacket thickness minimum	Outside diameter of cord, maximum	Dielectric withstanding voltage	Insulation resistance at 15.6°C, minimum	Capacitance, maximum
		Inch	Inch	Inch	Volts, rms	Megohms-1,000 ft.	Pf/ft.
WSL-1(20)	1	0.015	0.015	0.165	1,000	500	85
*WSL-1H(20)	1	.040	.025	.255	2,000	1,000	45
WSL-2(20)	2	.015	.032	.290	1,500	500	70
WSL-3(20)	3	.015	.032	.300	1,500	500	80
WSL-4(20)	4	.020	.035	.330	1,500	500	95
WSL-5(20)	5	.020	.040	.390	1,500	500	95
WSL-6(20)	6	.020	.040	.420	1,500	500	95
WSL-7(20)	7	.020	.068	.440	1,500	500	95

*Heavy duty, high voltage application.

3.2 Construction. Cords shall be constructed in the following manner. Each conductor shall be 20 AWG stranded, tinned-coated, copper wire covered with a separator and insulated with styrene butadiene rubber. (The separator may be omitted if a free-stripping insulating compound is used.) The insulated conductors shall be wound in a left-hand direction using fillers, if necessary, to form a compact round core. The lay shall be 10 to 16 times the pitch diameter of the cabled conductors. A closely woven binder or serving shall be applied over the cabled conductors, and a tinned-coated copper shield applied over the binder or serving. A separator shall be closely wound over the shield and a black styrene butadiene rubber jacket applied over the separator. The cord ends shall be sealed.

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3.3 Materials. The materials used in the fabrication of cords shall be as hereinafter specified.

3.3.1 Conductors. Conductors shall be 20 AWG, stranded, soft or drawn-and-annealed tinned-coated copper wire conforming to QQ-W-343, type B, class M.

3.3.2 Yarn. All yarn used for separators, servings, binders or fillers shall be synthetic fiber conforming to MIL-C-572, type P or CTA, or polyester yarn. The yarn used shall be clean, dry, and substantially free from all foreign particles, knots, lumps, or any substance that might impair the insulation of the wire.

3.3.3 Tape. Tape used for separators shall be polyester conforming to MIL-I-631, type G, form T_f.

3.3.4 Insulation. A styrene butadiene rubber insulating compound conforming to MIL-I-3930, type IS-L, shall be applied over the conductor or separator, when present, to a minimum thickness as specified in table I. The insulation shall be capable of being readily stripped from the conductor by standard methods, leaving the conductor clean for soldering. Insulated conductors shall be readily separable from each other.

3.3.5 Color coding. The conductors within the cord shall be clearly distinguishable by the color of the insulation. The coloring shall be solid, and may either extend through the thickness of the insulation or consist of a thin layer of colored compound applied over the insulation. In the latter case, the colored layer shall adhere firmly to the insulation wall. The color limits shall conform to MIL-STD-104, class 2. The color coding of the conductor insulation shall be in accordance with table II.

Table II. Color coding of conductor insulation.

No. of conductors	Color coding
1	Black
2	Black and white
3	Black, white and red
4	Black, white, red and green
5	Black, white, red, green and orange
6	Black, white, red, green, orange and blue
7	Black, white, red, green, orange, blue and brown

3.3.6 Shield. The shield shall be a closely woven braid of 34 AWG tinned-coated copper strands conforming to QQ-W-343, and shall be applied to provide coverage of not less than 85 percent of the construction underneath. (See 4.5.1.)

3.3.7 Jacket. A black styrene butadiene rubber jacket conforming to MIL-I-3930, type JS-L, shall be applied over the shield and separator to a minimum thickness as specified in table I.

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3.3.8 Outside diameter. The outside diameter of the completed cord shall not exceed the maximum dimension specified in table I.

3.4 Electrical requirements.

3.4.1 Dielectric withstanding voltage. The finished cord shall withstand for one minute the voltage specified in table I when tested in accordance with 4.5.2.1, and shall show no evidence of breakdown.

3.4.2 Insulation resistance. The insulation resistance shall be not less than the value specified in table I when tested in accordance with 4.5.2.2.

3.4.3 Conductor resistance. The direct current (dc) resistance of each conductor in a finished cord shall not exceed 11.57 ohms (max.)/1,000 feet at 20°C when tested in accordance with 4.5.2.3.

3.4.4 Capacitance. The capacitance of the finished cord shall not exceed the value specified in table I when tested in accordance with 4.5.2.4.

3.5 Physical properties.

3.5.1 Cold bend. Neither the jacket nor the insulation shall show evidence of cracks, flaws, or other damage when tested in accordance with 4.5.3.1 at a temperature of -55° ±2°C.

3.6 Lengths. Unless otherwise specified (see 6.2), the cord shall be furnished wound on spools or reels in continuous lengths of not less than 250 feet.

3.7 Sealing of cord ends. To prevent moisture from entering the finished cord, both ends of each shipping length of all cords shall be dipped in a sealer which shall completely seal the ends. No treatment is required over that part of the cord covered by the jacket.

3.8 Workmanship. Cords shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques. The cords shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection equipment and facilities. Inspection equipment and facilities shall be established and maintained in accordance with MIL-C-45662.

4.2 Classification of inspections. The examinations and tests of cords are classified as follows:

(a) Materials inspection. (See 4.3.)

(b) Quality conformance inspection. (See 4.4.)

1. Inspection of product for delivery. (See 4.4.1.)

2. Inspection of preparation for delivery. (See 4.6.)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table III, used in fabricating the designated cord type (see 3.1), are in accordance with the applicable referenced specification or requirements prior to such fabrication.

Table III. Materials inspection.

Material	Requirement paragraph	Applicable specification
Conductors	3.3.1	QQ-W-343
Yarn	3.3.2	MIL-C-572
Tape	3.3.3	MIL-I-631
Insulation	3.3.4	MIL-I-3930
Shield	3.3.6	QQ-W-343
Jacket	3.3.7	MIL-I-3930

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and C inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all cords of the same type (see 3.1) produced under essentially the same conditions and submitted for inspection at the one time.

4.4.1.2 Unit of product. Unless otherwise specified in the contract or order, the unit of product for purposes of sampling shall be each continuous length of cord contained on a reel or spool. (See 3.6.)

4.4.1.3 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan for the presented lot.

4.4.1.4 Selection of sample units. Sample units for inspection shall be taken from each unit of product which forms part of the sample. A sample unit is defined as a length of cord drawn from a unit of product.

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4.4.1.5 Test specimen. A test specimen may be the entire sample unit (length of cord) or any portion of the sample unit which is to be tested.

4.4.2 Group A inspection. Group A inspection shall include the examinations and tests of table IV, subgroups I and II. Major and minor defects shall be as defined in MIL-STD-105.

4.4.2.1 Subgroup I. For subgroup I, the acceptable quality level (AQL) shall be as specified in table IV and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup I tests may be performed in any order.

4.4.2.2 Subgroup II. For subgroup II, every length of cord in the inspection lot shall be subjected to the tests. Subgroup II tests shall be performed in the order shown in table IV. The entire lot shall be rejected if more than 3 defects per 1000 feet in a lot are found. All electrical defects are considered major.

Table IV. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Subgroup I				
Visual and dimensional				
Construction	3.2	4.5.1	1.0% for the subgroup	4.0% for the subgroup
Conductors	3.3.1			
Yarn	3.3.2			
Tape	3.3.3			
Insulation	3.3.4			
Color coding	3.3.5			
Shield	3.3.6			
Jacket	3.3.7			
Outside diameter	3.3.8			
Lengths	3.6			
Sealing of cord ends	3.7			
Workmanship	3.8			
Subgroup II				
Electrical				
Dielectric withstanding voltage	3.4.1	4.5.2.1	See 4.4.2.2	
Insulation resistance	3.4.2	4.5.2.2		
Conductor resistance	3.4.3	4.5.2.3		
Capacitance	3.4.4	4.5.2.4		

4.4.2.3 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

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4.4.3 Group B inspection. Not applicable.

4.4.4 Group C inspection. This inspection shall consist of the cold bend tests specified in 4.5.3.1 and shall be performed periodically as indicated in 4.4.4.1. Samples shall be selected from lots that have passed group A inspection.

4.4.4.1 Sampling for inspection. One sample shall be selected per each 5,000 feet of cord or fraction thereof, not to exceed a total of five samples per order. Samples shall be selected periodically through the life of the contract to assure representative results.

4.4.4.2 Noncompliance. No failures shall be allowed in group C inspection. If a sample unit fails to pass group C inspection, the supplier shall take corrective action on the material or process or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection that the original sample failed, at the option of the government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5 Test methods.

4.5.1 Visual and dimensional examination. The specimens of finished cords selected for the group A, subgroup I inspection (see 4.4.2.1) shall be examined for conformity to those parts of requirements 3.2 to 3.3.8 inclusive, 3.6, 3.7, and 3.8 which can be determined by visual and dimensional examination. Conformity to the shield coverage requirement (see 3.3.6) shall be determined as specified in FED-STD-228, Method 8121. Visual and dimensional defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105.

4.5.2 Electrical tests.

4.5.2.1 Dielectric withstanding voltage. (See 3.4.1.) The finished cord shall be tested as specified in FED-STD-228, Method 6111, except for the following:

(a) The test shall be performed on finished cord only.

(b) The immersion period shall be not less than 9 hours.

(c) One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with the shield and the water.

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4.5.2.2 Insulation resistance. (See 3.4.2) The insulation resistance of the finished cord shall be determined as specified in FED-STD-228, Method 6031 except for the following:

(a) The test shall be performed on finished cord only.

(b) The immersion period shall be not less than 9 hours.

(c) The test voltage shall be not less than 100 volts nor more than 500 volts dc.

(d) The polarity of the conductor shall be maintained negative with respect to the water. One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with one another and the shield, and the water.

(e) If the measurement is made at a temperature lower than 15.6°C, the supplier shall correct the measured value of insulation resistance to the resistance at 15.6°C.

4.5.2.3 Conductor resistance. The direct current (dc) resistance of each conductor in a finished cord shall be determined in accordance with FED-STD-228, Method 6021, and shall meet the requirements of 3.4.3.

4.5.2.4 Capacitance. The capacitance per foot of cord between each conductor and all other conductors connected together with the shield shall be measured at a frequency of 1000 Hz by using an alternating current balanced bridge method accurate to within 1/4 of 1 percent. (See 3.4.4.)

4.5.3 Physical tests.

4.5.3.1 Cold bend. (See 3.5.1.)

4.5.3.1.1 Specimens. One specimen shall be cut from each sample unit. Each specimen so cut shall be divided into two parts, one for checking the cord as a whole, and the other for checking the insulation apart from the cord.

4.5.3.1.2 Procedure. The specimens selected for checking cord as a whole shall be attached to a mandrel of the proper size as specified in 4.5.3.1.3. The specimens selected for checking the insulation apart from the cord shall have the jacket removed, and each insulated conductor therefrom shall be attached to a mandrel 0.062 inch in diameter. The specimens shall be suspended vertically, with their lower ends weighted sufficiently to keep the specimens taut and to permit bending them without handling. The mandrel and specimen shall be placed for at least 20 hours in a cold chamber at a temperature of -55° ±2°C and, while at this temperature, bent for five close turns around the

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mandrel at the rate of 15 ±3 turns per minute. After the test has been completed, the jacket on the specimens of the cord shall be examined through a magnifying glass of at least 3-diameter magnification. The jacket shall then be removed so that the insulation can be examined. The insulation on all specimens shall be examined for cracks with the magnifying glass.

4.5.3.1.3 Mandrels for cord. The mandrel for testing the cord as a whole shall be selected from the following list of standard size mandrels.

Mandrel diameters (inches)
0.68
0.84
1.05
1.31
1.66
1.90

The size selected shall be the largest size which does not exceed three times the specified maximum overall diameter of the cord. (See 3.1.)

4.6 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking shall be in accordance with MIL-C-12000. (See 6.2.)

6. NOTES

6.1 Intended use. The cords covered by this specification are intended for use as patch cords for telephone switchboards and for use with microphones, headsets, handsets, etc., where flexibility is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type designation of cord required. (See 3.1.)

(c) Length of finished cords to be delivered. (See 3.6.)

(d) Level of packaging and level of packing required. (See 5.1.)

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6.2.1 Indirect shipments. The preservation, packaging, packing and marking specified in section 5 apply only to direct purchases by or direct shipments to the government, and are not intended to apply to contracts or orders between the supplier and prime contractor.

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Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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